

**UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 396 – SENIOR DESIGN I
Fall 2019**

Lecture:

Section 1: T/Th: 9:45am – 11am: 271 EIB

Section 2: T/Th: 1pm-2:15pm: 124 EIB

No Classes:

November 28nd (Thanksgiving Break)

Office Hours:

11am-1pm Tuesdays and Thursdays (Dr. Bilgin) 144 EIB

Monday 12-1pm with TA

Instructor:

Dr. Betul Bilgin, bbilgin@uic.edu

Teach Asst:

Aaditya Pendse, apends2@uic.edu

Prerequisites:

CHE 312, CHE 313 and CHE 312

Text:

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012. (Textbook website: booksite.elsevier.com/Towler) (required)

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009 (as a reference)

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker (printed book ISBN: 9781932780093 / retail price: \$39 or e-book ISBN: 9781932780109 / retail price \$19) (as a reference)

Course Description:

This is an introduction class to modern process design and development, engineering economics and report writing. It also covers design and cost of equipment related to materials handling to heat transfer and reactors.

Course Objectives:

- Provide a real-world perspective on process design
- Introduce design and selection of equipment
- Give insight into practical constraints that govern design decisions
- Help you translate knowledge into practice & start solving real problems
- Improve presentation, analytical thinking and team working skills

Course Outline:

Date	Chapter	Assignment
August 27 th 2019	Syllabus Development	
August 29 th 2019	Chapter 1 (Introduction)	
September 3 rd 2019	Chapter 1 (Introduction)	
September 5 th 2019	Chapter 2 (Flowsheet development)	
September 10 th 2019	Chapter 2 (Flowsheet development)	
September 12 th 2019	Chapter 2 (Flowsheet development)	
September 17 th 2019	Chapter 2 (Flowsheet development)	1 st Assignment Due
September 19 th 2019	Presentation 1	
September 24 th 2019	Presentation 1	Quiz 1
September 26 th 2019	Chapter 3 (Utilities and Energy Recovery)	
October 1 st 2019	Chapter 3 (Utilities and Energy Recovery)	
October 3 rd 2019	Chapter 4 (Process simulation)	
October 8 th 2019	Chapter 4 (Process simulation)	
October 10 th 2019	Chapter 5 (Process control)	2 nd Assignment is Due 10/12/2019
October 15 th 2019	Chapter 5 (Process control)	Quiz 2

October 17 th 2019	Chapter 6 (Materials of construction)	
October 22 nd 2019	Chapter 7 (Capital cost)	3 rd Assignment Due
October 24 th 2019	Chapter 7 (Capital cost)	Quiz 3
October 29 th 2019	Chapter 8 (Operating cost)	
October 31 st 2019	Chapter 8 (Operating cost)	
November 5 th 2019	Chapter 9 (Economic Analysis)	
November 7 th 2019	Chapter 9 (Economic Analysis)	4 th Assignment Due Presentation 2
November 12 th 2019	Chapter 9 (Economic Analysis)	Quiz 4 Presentation 2
November 14 th 2019	Chapter 10 (Process Hazard, Professional ethics)	
November 19 th 2019	Chapter 10 (Process Hazard, Professional ethics)	5 th Assignment Due
November 21 st 2019	Professional Ethics Presentations	
November 26 th 2019	Professional Ethics Presentations	
November 28 th 2019	NO Class	
December 3 rd 2019	Chapter 11 (General Site Consideration)	6 th Assignment Due Quiz 5
December 5 th 2019	Review	Final Project Due

Class Structure:

In a typical class,

- You are required to read the topic before the class
 - During half of class time, I will present key points of the topic
 - After presentation of the topic, I will give you a small project design that you will be working in the class with a group of 5.
- In-class projects are expected to be the product of team efforts. Each member of the team is expected to contribute equally to the development of the design. In- class projects will not be

graded. Feedback will be provided by the instructor.

Homework / mini-project assignments:

There will be 6 biweekly assignments in fall semester. For each assignment, you will be working with a group of total four students who will be randomly selected. Therefore, you will have the opportunity to work with different groups of people and to practice team work. I expect that all the members of the group contribute equally to the group's project. Homework due dates are listed on the syllabus. They are due midnight on due date. Late homework will be accepted up to 24 hours after due date for 50% credit. Solutions to homework will be presented in the class by students. Instructor will be randomly selecting which group will present which homework question. You should be prepared to present any question of homework. Also, peer evaluations will be very important. Your peers in particular homework will be evaluating your performance. Those evaluations will be graded and it will affect your homework grade. Peer evaluations will be done on catme.org and average evaluation scores of group members will be calculated. If your peer evaluation average is less than 75%, your grade will be calculated by multiplying group grade by peer evaluation average percentage.

Final Project/ Proposals (written):

A major design project will be assigned to each group of students at the end of October. Student groups will include 4 members. Each group will have their own design project. Each group is required to submit a proposal at the end of semester. Proposal should include background material about the project, explanation of your design objective and process description. Written proposal will be due the end of semester.

Exams:

There will be NO exam for this course.

Presentations:

Students are required to give two oral presentations throughout the course. First presentation will be based on HW 1 topic for 10 min. Second presentation will be group of three for 10 min. Topic choices for second presentation will be posted. Presentation should be prepared for general public audience as well as for chemical engineers. Presentations will be graded based on the design of the presentation, how you form the story and how you tell the story. Presentations will be graded by the instructor. Classmates will be providing feedback.

Quizzes :

There will be 5 individual quizzes throughout the course. Quizzes will be based on HW and presentations. Quizzes are for assessing individual learning.

Attendance/Participation:

Students are expected to attend all classes, and attendance is incorporated as 10% of the final grade. Students are allowed two excused absences and one unexcused absence. If absences exceed these limits, the student will receive a zero for attendance. Participation to class is also important. It will be incorporated to attendance grade. In-class assignments will be graded as participation.

SaChE safety modules:

Students are expected to complete SaChE safety modules online and submit their certificates of completion to the instructor. There are only two relevant SaChE modules for ChE 396:

- Inherently safer design
- Safety in the process industries

Grading Policy:

Course grades will be weighted according to the following grading formula.

Attendance/ Participation:10% (in-class assignments will be incorporated to participation)
Homework/Mini-Project Assignments:.....45% (6 of them, group of 4, written; individual grade will be calculated based on multiplication of group grade with peer evaluation percentage)
Oral Presentations:20% (2 times, first one is based on HW 1 so group of 4, second one is group of 3)
Quizzes:10% (5 of them, individual, based on HWs, lowest will be dropped)
Final Project/ Proposal:..... 15% (written only)

The expectation is that all students will do well in this course if they actively participate in the course, do assignments on time and keep up with course materials.

Most relevant student outcomes for this course:

SO	Description
1	Ability to apply knowledge of mathematics, science and engineering
2	Ability to identify, formulate and solve engineering problems
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice
7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability

8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
9	Knowledge of contemporary issues
10	Recognition of the need for and ability to engage in life-long learning
11	Ability to function on multidisciplinary teams
12	Ability to communicate effectively
13	Understanding of professional and ethical responsibilities